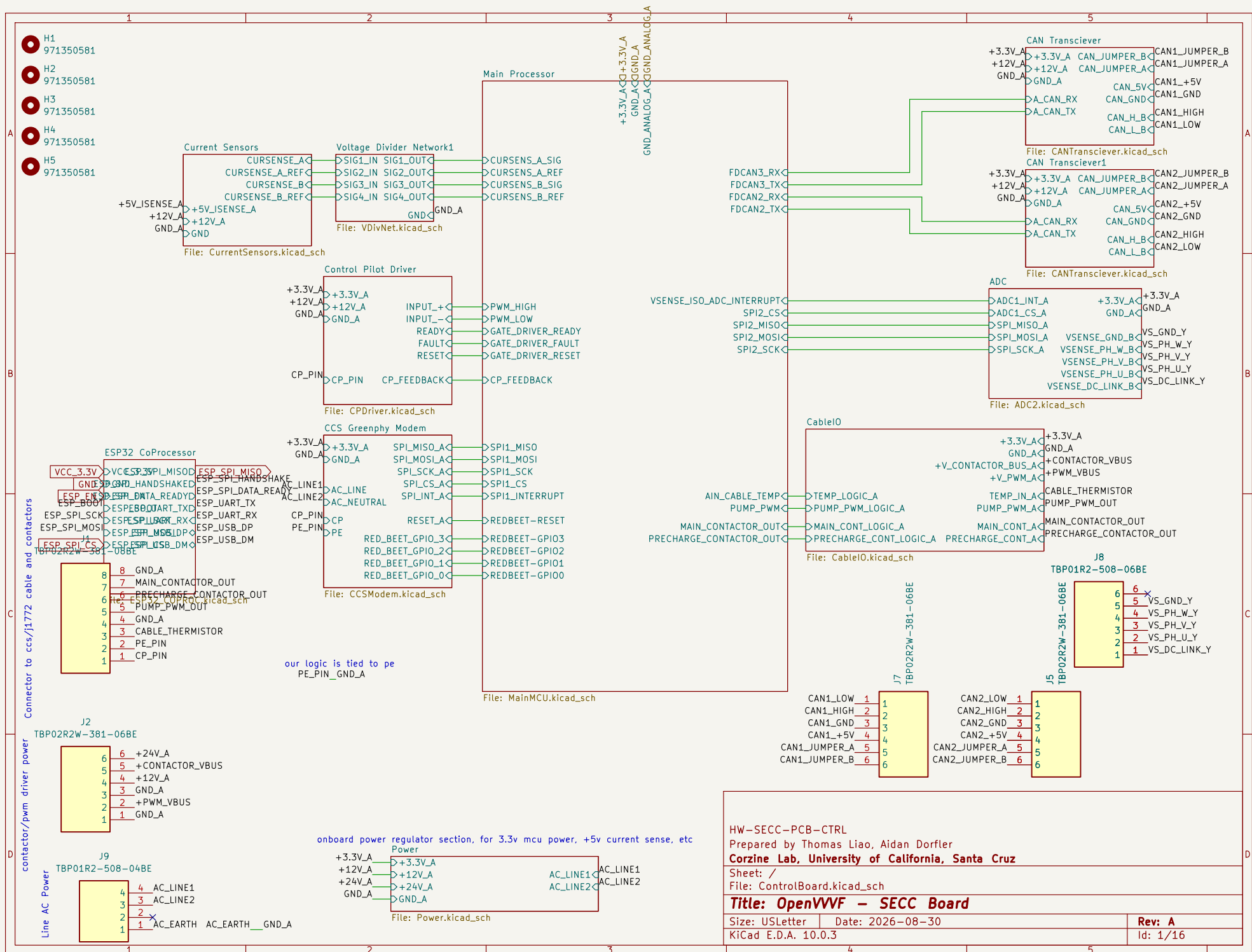
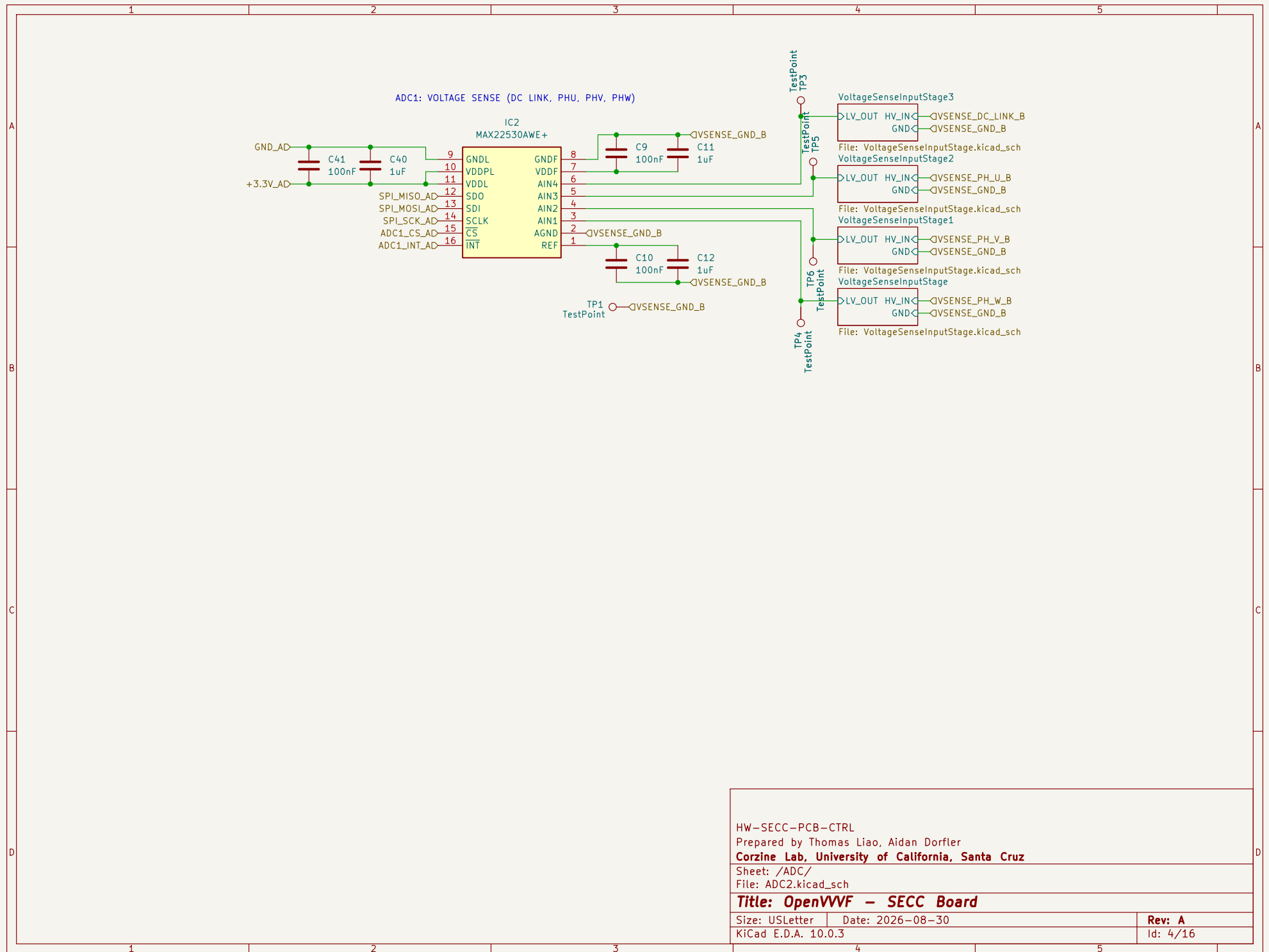






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Sheet: /ADC/

File: ADC2.kicad_sch

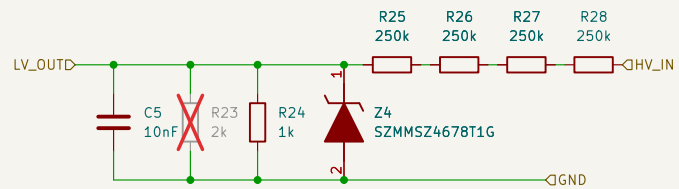
Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30

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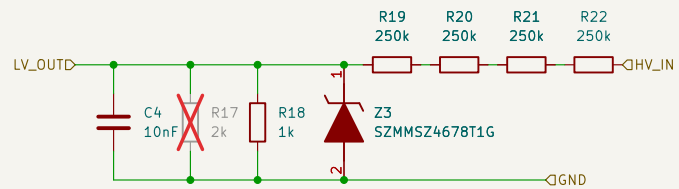
ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch

Title: OpenVWF - SECC Board

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Id: 2/16



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

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Sheet: /ADC/VoltageSenseInputStage1/

File: VoltageSenseInputStage.kicad_sch

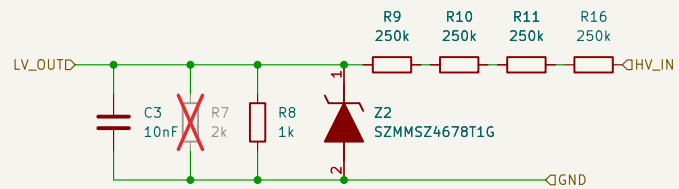
Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30

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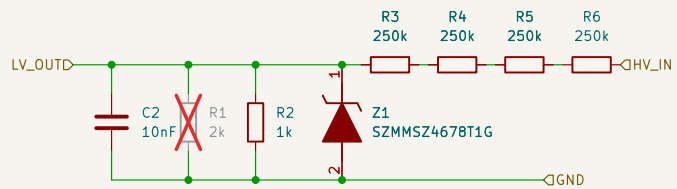
ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
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Sheet: /ADC/VoltageSenseInputStage2/
File: VoltageSenseInputStage.kicad_sch

Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30
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Id: 5/16



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-SECC-PCB-CTRL

Prepared by Thomas Liao, Aidan Dorfler

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Sheet: /ADC/VoltageSenseInputStage3/

File: VoltageSenseInputStage.kicad_sch

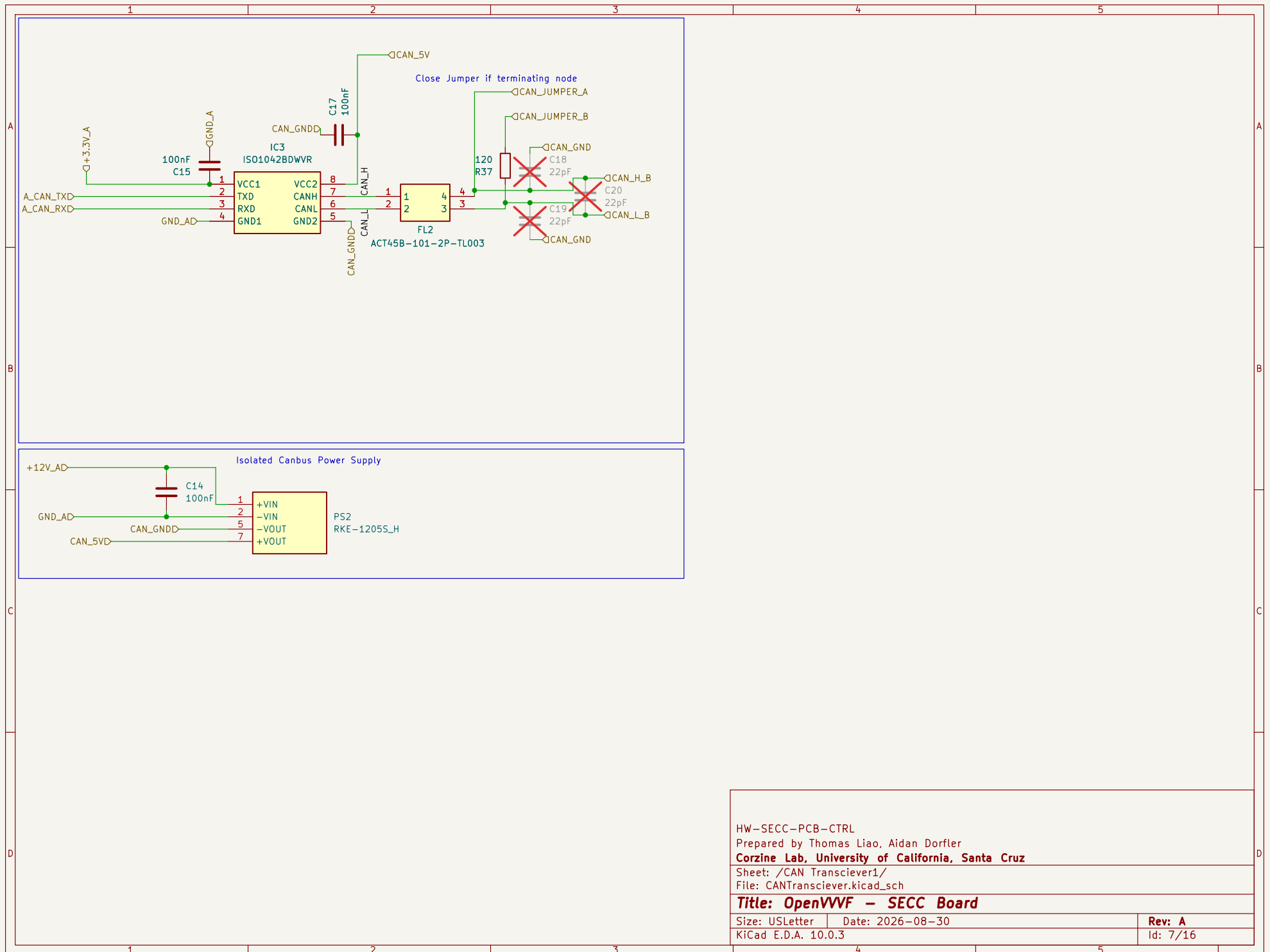
Title: OpenVWF - SECC Board

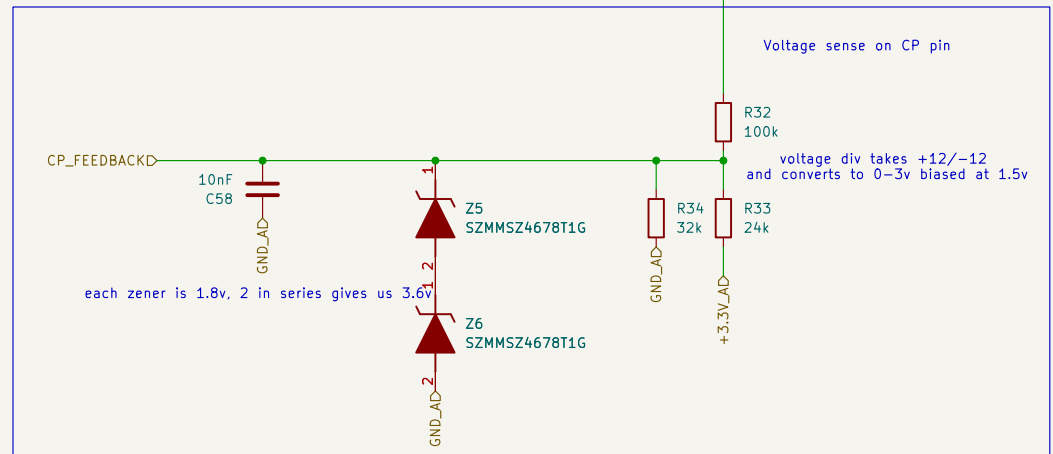
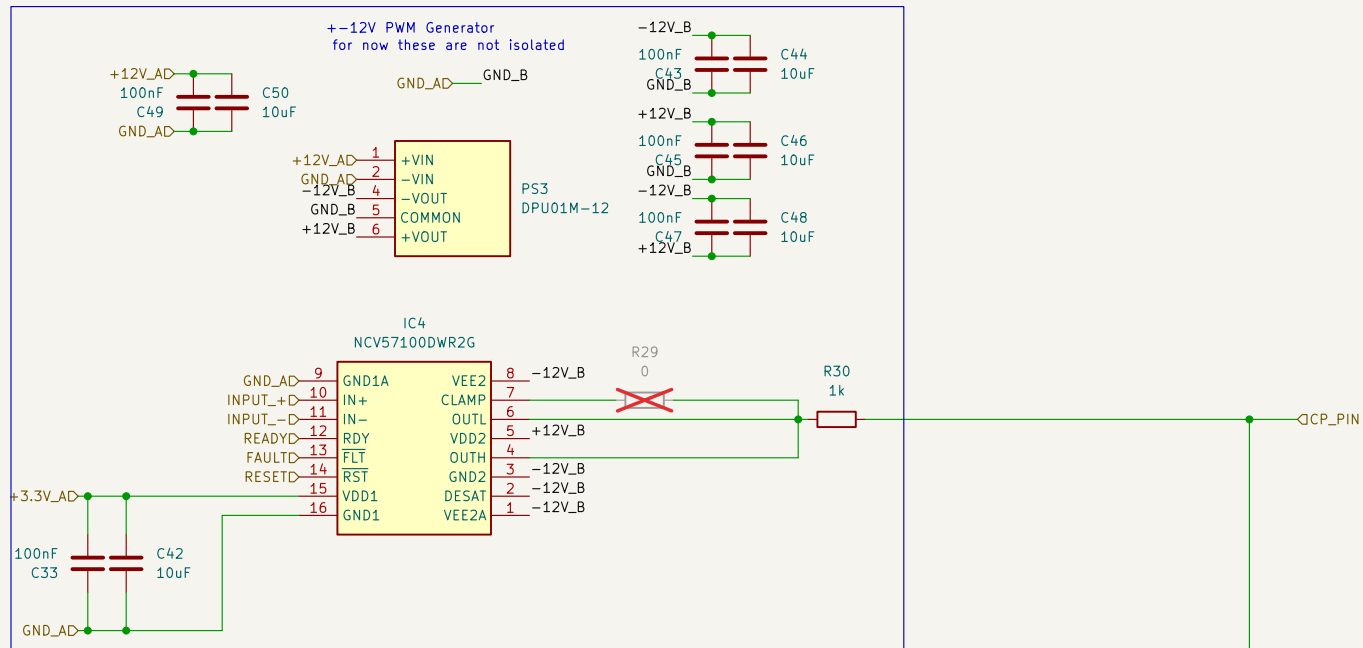
Size: USLetter Date: 2026-08-30

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Id: 6/16





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Sheet: /Control Pilot Driver/

File: CPDriver.kicad_sch

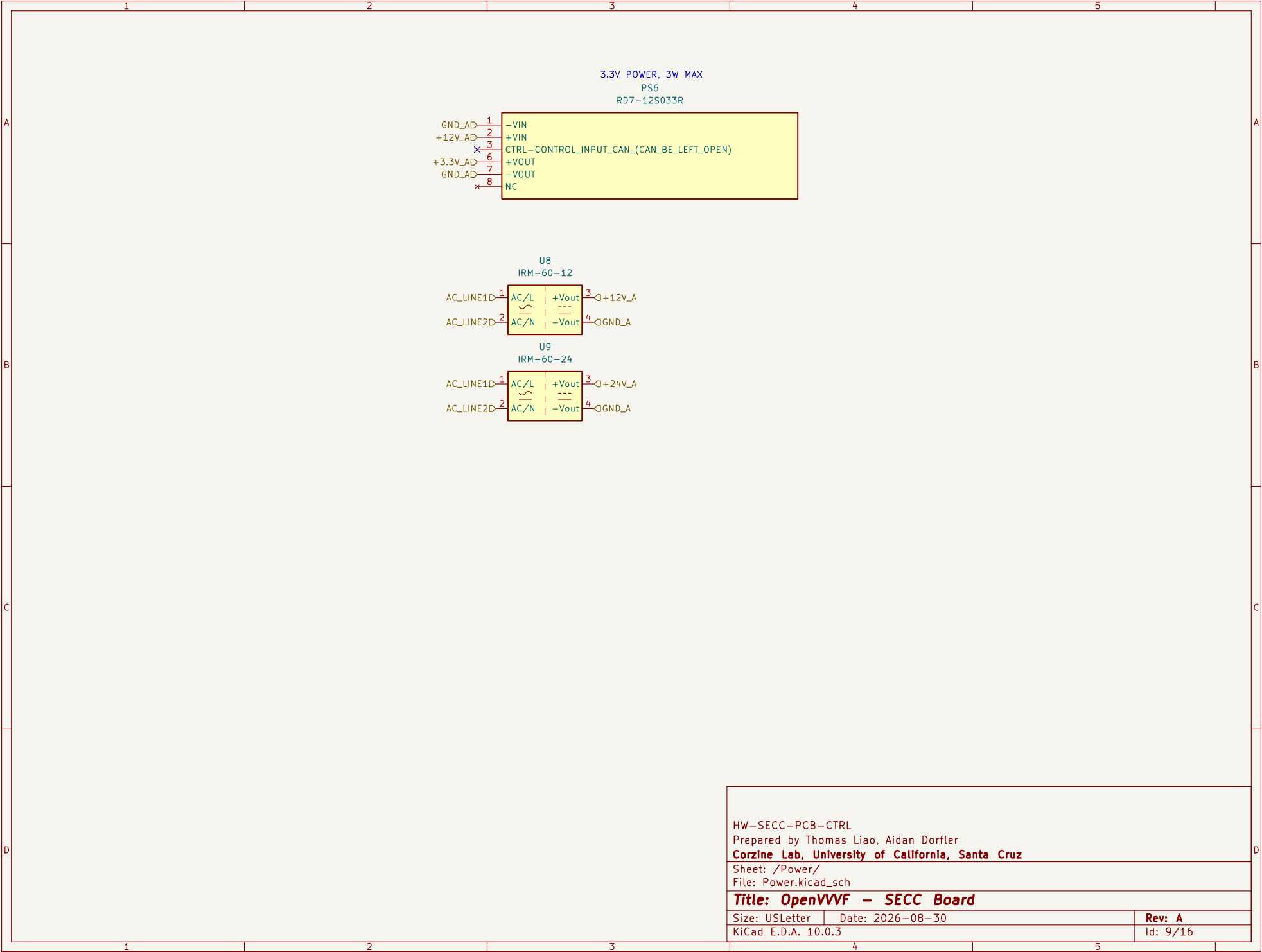
Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30

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Rev: A

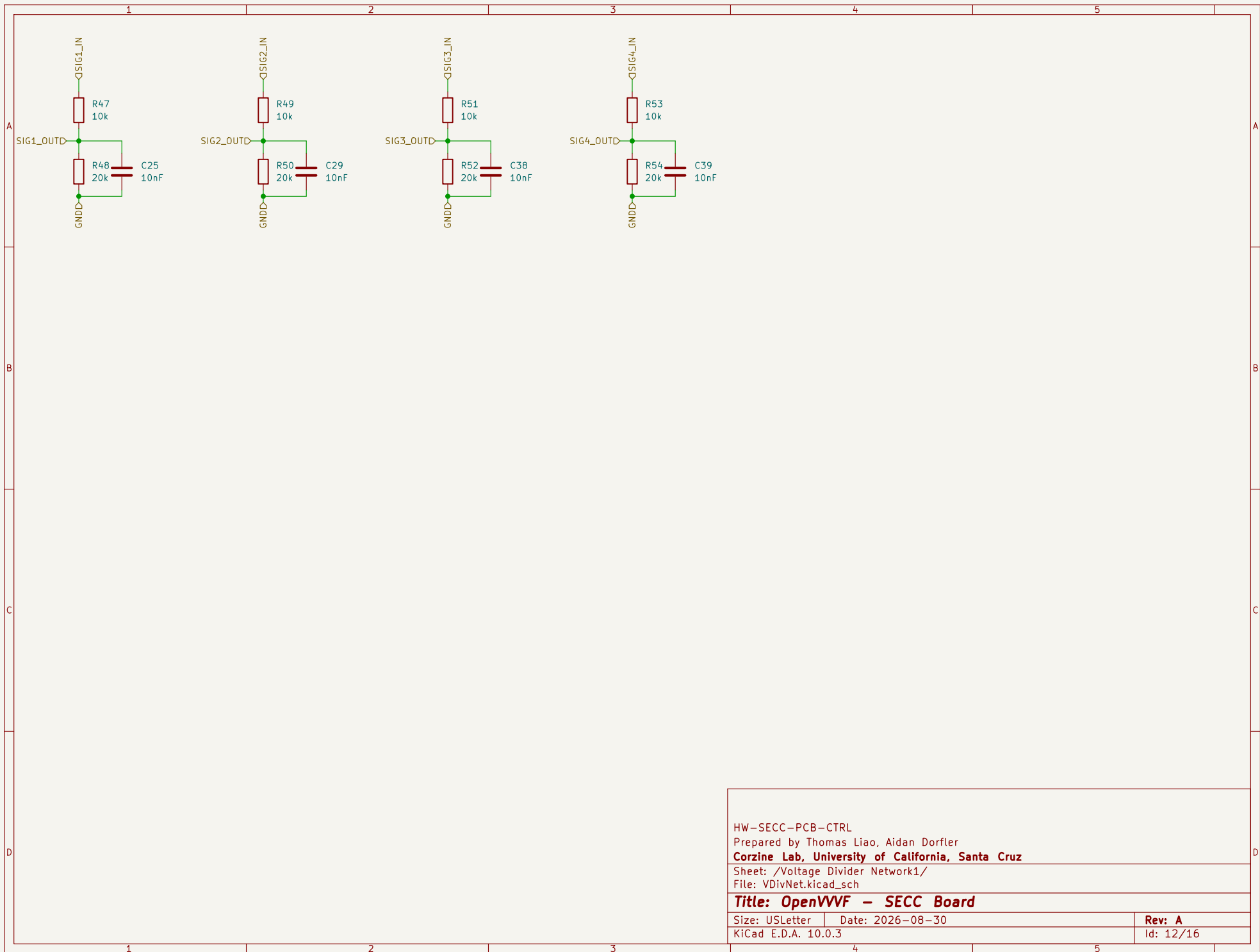
Id: 8/16



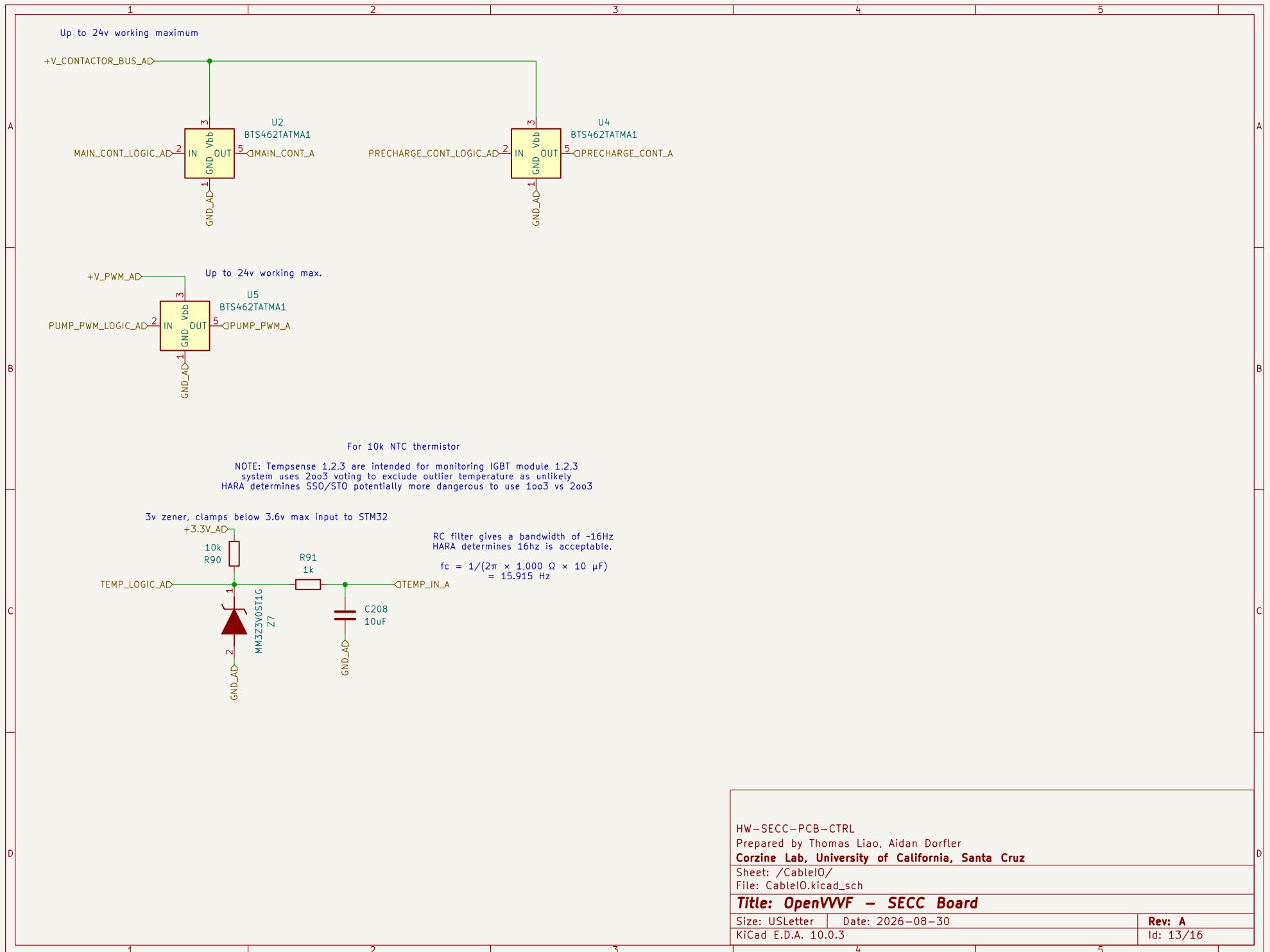
HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
Corzine Lab, University of California, Santa Cruz
Sheet: /Power/
File: Power.kicad_sch

Title: OpenVWF - SECC Board

Size: USLetter	Date: 2026-08-30	Rev: A
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Sheet: /Voltage Divider Network1/
File: VDivNet.kicad_sch
Title: OpenVWF - SECC Board
Size: USLetter Date: 2026-08-30
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Sheet: /CableIO/

File: CableIO.kicad_sch

Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30

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Rev: A

Id: 13/16

The schematic shows the electrical connections for an ESP32-C3-WROOM-02 module (U10) on a PCB. The module is a yellow rectangle with pins numbered 1 through 19. The connections are as follows:

- Power and Ground:**
 - VCC_3.3VD is connected to pin 1 (EN) and pin 3 (IO20/RXD).
 - GNDD is connected to pin 2 (EN), pin 18 (IO0), pin 19 (IO19), and a ground symbol.
 - A PWR_FLAG is connected to the VCC_3.3VD line.
- Reset and Timing:**
 - C1 (10uF, 25V) is connected to VCC_3.3VD and GNDD.
 - C6 (0.1uF, 25V) is connected to VCC_3.3VD and GNDD.
 - R12 (10k, 1%) is connected to VCC_3.3VD and GNDD.
 - C7 (1uF, 16V) is connected to VCC_3.3VD and ESP_SPI_MISO.
 - R14 (10k, 1%) is connected to VCC_3.3VD and ESP_SPI_MISO.
 - R13 (10k, 1%) is connected to VCC_3.3VD and ESP_SPI_SCKD.
 - R15 (10k, 1%) is connected to VCC_3.3VD and ESP_SPI_MOSID.
- Signal Connections:**
 - ESP_END is connected to pin 2 (EN).
 - ESP_SPI_MISO is connected to pin 18 (IO0).
 - ESP_SPI_HANDSHAKE is connected to pin 17 (IO1).
 - ESP_SPI_DATA_READY is connected to pin 16 (IO2).
 - ESP_SPI_SCKD is connected to pin 5 (IO5).
 - ESP_SPI_MOSID is connected to pin 6 (IO6).
 - ESP_BOOTD is connected to pin 7 (IO8).
 - ESP_SPI_CSD is connected to pin 10 (IO10).
 - ESP_USB_DM is connected to pin 13 (IO18).
 - ESP_USB_DP is connected to pin 14 (IO19).
 - IO21/TXD is connected to pin 12 (IO12).
 - IO20/RXD is connected to pin 11 (IO11).

Annotations and notes:

- At the top left: "A peaks (Wi-Fi TX). Place C6 closer to pin 1 than C1. mode, up to 60 MHz. Keep traces short, length-matched. not available (in-package flash)."
- Next to the module: "cross to H7 UART: ESP_UART_TX -> H7_RX, ESP_UART_RX -> H7_TX. H7 idles its TX (ESP_UART_RX) high. Enter download mode: ESP_BOOT low + pulse ESP_EN low."
- Below the module: "H7 drives open-drain or push-pull-LOW only. EN is shared with the R12/C7 power-on-reset RC, so the H7 must never source current into it (a push-pull high would override the RC delay and fight any pulse on the net). Reserved for future USB connector (native USB-Serial/JTAG flashing/debug)."

Bottom right information:

HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
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Sheet: /ESP32 CoProcessor/
File: ESP32_COPROC.kicad_sch
Title: OpenVVF - SECC Board
Size: USLetter Date: 2026-08-30 Rev: A
KiCad E.D.A. 10.0.3 Id: 15/16

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